Worksheet 1.6 — Logarithms

Mathematics: Analysis & Approaches SL (IB) · Unit 1: Number and Algebra

The Big Ideas

Logarithms are the inverse of exponentials: $\log_b(x) = y \iff b^y = x$.

- $\log_b(mn) = \log_b m + \log_b n$, $\log_b\left(\dfrac{m}{n}\right) = \log_b m - \log_b n$, $\log_b(m^k) = k\log_b m$.
- Change of base: $\log_b a = \dfrac{\ln a}{\ln b}$.
- Use logarithms to solve exponential equations with unmatched bases.

Worked Examples

Example 1 — Evaluate directly

Evaluate $\log_3(81)$.

Solution. $3^{?} = 81 = 3^4$.

Conclusion: 4.

Example 2 — Solve via logarithms

Solve $6^x = 50$ to 3 decimal places.

Solution. Take ln of both sides: $x\ln 6 = \ln 50 \Rightarrow x = \dfrac{\ln 50}{\ln 6}$.

Conclusion: $x \approx 2.184$.

Example 3 — Simplify using log laws

Simplify $\log_2(20) + \log_2(4) - \log_2(5)$.

Solution. $= \log_2\left(\dfrac{20 \times 4}{5}\right) = \log_2(16)$.

Conclusion: 4.

Example 4 — Solve with a domain check

Solve $\log_3(x) + \log_3(x-2) = 1$, checking the domain.

Solution. **Step 1:** $\log_3\left(x(x-2)\right) = 1 \Rightarrow x(x-2) = 3 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x-3)(x+1) = 0$.

Step 2: $x = 3$ or $x = -1$; but $\log_3(x)$ requires $x > 0$, so $x = -1$ is rejected.

Conclusion: $x = 3$.

Example 5 — Change of base

Evaluate $\log_7(50)$ to 3 decimal places using change of base.

Solution. $\log_7(50) = \dfrac{\ln 50}{\ln 7}$.

Conclusion: ≈ 2.010.

Example 6 — Solve for an exponent using change of base

Solve $3^{x+1} = 20$ to 3 decimal places.

Solution. Take ln: $(x+1)\ln 3 = \ln 20 \Rightarrow x + 1 = \dfrac{\ln 20}{\ln 3}$.

Conclusion: $x \approx 1.727$.

Example 7 — Application — half-life

A substance decays according to $A = A_0(0.5)^{t/8}$ (8-day half-life). Find, to the nearest day, when only 20% remains.

Solution. $0.2 = 0.5^{t/8} \Rightarrow \ln 0.2 = \dfrac{t}{8}\ln 0.5 \Rightarrow t = 8 \cdot \dfrac{\ln 0.2}{\ln 0.5}$.

Conclusion: $t \approx 19$ days.

Example 8 — Combine multiple log laws

Simplify $3\log_2(x) - \log_2(x^2) + \log_2(4)$ as a single logarithm.

Solution. $= \log_2(x^3) - \log_2(x^2) + \log_2(4) = \log_2\left(\dfrac{4x^3}{x^2}\right)$.

Conclusion: $\log_2(4x)$.

Example 9 — Solve a log equation reducing to a quadratic

Solve $(\log_2 x)^2 - 3\log_2 x + 2 = 0$ (let $u = \log_2 x$).

Solution. $u^2 - 3u + 2 = 0 \Rightarrow (u-1)(u-2) = 0 \Rightarrow u = 1$ or $u = 2$.

Conclusion: $\log_2 x = 1 \Rightarrow x = 2$; $\log_2 x = 2 \Rightarrow x = 4$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Evaluate $\log_5(125)$.

Question 2

Solve $4^x = 70$ to 3 decimal places.

Question 3

Simplify $\log_4(48) - \log_4(3)$.

Question 4

Solve $\log_2(x) + \log_2(x-6) = 4$, checking the domain.

Question 5

An investment is $A = 800(1.05)^t$. Find, to the nearest year, when it first exceeds \$1500.

Question 6

Simplify $2\log_5(3) + \log_5(4)$ as a single logarithm, then evaluate it.

Question 7

The Richter scale uses $M = \log_{10}(I/I_0)$. Find M if an earthquake is 12,000 times as intense as I_0.

Question 8

Evaluate $\log_9(200)$ to 3 decimal places using change of base.

Question 9

A substance decays as $A = A_0(0.5)^{t/5}$; find, to the nearest day, when 30% remains.

Question 10

Solve $(\log_3 x)^2 - 4\log_3 x + 3 = 0$.

Question 11 — Challenge

Solve $\log_2(x+3) - \log_2(x-1) = 3$, checking the domain.

Question 12 — Challenge

Solve $2\ln(x) = \ln(x+6) + \ln(4)$ for $x > 0$.

Question 13 — Challenge

Solve $5^{2x+1} = 3^x$ to 3 decimal places by taking ln of both sides.

Question Recap & Answer Key

Question Recap

1. Evaluate $\log_5(125)$.
2. Solve $4^x = 70$ to 3 decimal places.
3. Simplify $\log_4(48) - \log_4(3)$.
4. Solve $\log_2(x) + \log_2(x-6) = 4$, checking the domain.
5. An investment is $A = 800(1.05)^t$. Find, to the nearest year, when it first exceeds \$1500.
6. Simplify $2\log_5(3) + \log_5(4)$ as a single logarithm, then evaluate it.
7. The Richter scale uses $M = \log_{10}(I/I_0)$. Find M if an earthquake is 12,000 times as intense as I_0.
8. Evaluate $\log_9(200)$ to 3 decimal places using change of base.
9. A substance decays as $A = A_0(0.5)^{t/5}$; find, to the nearest day, when 30% remains.
10. Solve $(\log_3 x)^2 - 4\log_3 x + 3 = 0$.
11. Solve $\log_2(x+3) - \log_2(x-1) = 3$, checking the domain.
12. Solve $2\ln(x) = \ln(x+6) + \ln(4)$ for $x > 0$.
13. Solve $5^{2x+1} = 3^x$ to 3 decimal places by taking ln of both sides.

✓ Answer Key

1. 3
2. $x = \dfrac{\ln 70}{\ln 4} \approx 3.075$
3. $\log_4(16) = 2$
4. $x(x-6) = 16 \Rightarrow x^2 - 6x - 16 = 0 \Rightarrow (x-8)(x+2) = 0 \Rightarrow x = 8$ (rejecting $x = -2$)
5. $1.05^t > 1.875 \Rightarrow t > \dfrac{\ln 1.875}{\ln 1.05} \approx 12.9$, so $t = 13$ years
6. $\log_5(36)$; $5^{2.226\ldots} = 36$, so value ≈ 2.226
7. $M = \log_{10}(12000) \approx 4.079$
8. $\dfrac{\ln 200}{\ln 9} \approx 2.415$
9. $0.3 = 0.5^{t/5} \Rightarrow t = 5 \cdot \dfrac{\ln 0.3}{\ln 0.5} \approx 9$ days
10. let $u = \log_3 x$: $(u-1)(u-3) = 0$; $x = 3$ or $x = 27$
11. $\log_2\left(\frac{x+3}{x-1}\right) = 3 \Rightarrow \frac{x+3}{x-1} = 8 \Rightarrow x + 3 = 8x - 8 \Rightarrow x = \frac{11}{7}$ (valid, since > 1)
12. $\ln(x^2) = \ln(4(x+6)) \Rightarrow x^2 = 4x + 24 \Rightarrow x^2 - 4x - 24 = 0 \Rightarrow x = \dfrac{4 \pm \sqrt{112}}{2} = 2 \pm 2\sqrt{7}$; reject the negative root, so $x = 2 + 2\sqrt{7}$
13. $(2x+1)\ln 5 = x\ln 3 \Rightarrow 2x\ln 5 + \ln 5 = x\ln 3 \Rightarrow x(2\ln 5 - \ln 3) = -\ln 5 \Rightarrow x = \dfrac{-\ln 5}{2\ln 5 - \ln 3} \approx -1.007$